Joint SPX/VIX calibration of the fBm-bank ESN with bounded readouts and a modulated rough bank

At the optimal architecture of the grid-search protocol — market of 19 February 2026

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Abstract

We calibrate the fBm-bank echo-state-network volatility model with bounded readouts and a modulated rough bank jointly to the SPX and VIX option markets of 19 February 2026. The architecture is fixed at the optimum of the grid-search protocol:

- rough bank $M = 18$, $H = 0.058$, rates $[10^{-3}, 232]$;
- leverage window $\Theta \approx 37$ days;
- feedback bank $J = 2$;
- modulation $\lambda_\Gamma = 2.47$, $\Lambda = 4.63$, $a_J = 1.06$;
- readouts with $(N_c, N_d, N_e) = (5, 4, 1)$ pieces.

The volatility is

$$\sigma_t = \tilde{\sigma}(t)\, e^{\psi(\zeta_t)}\, F(P_t), \qquad \tilde{\sigma}(t) = ae^{-bt} + c(1 - e^{-bt}).$$

SPX options are priced by Monte Carlo, and the VIX with a positive estimator (analytic conditional-variance base times a regressed correction). The jumps of Γ are drawn with stratified counts.

The 17 parameters (readouts, term structure, ϱ, today's state) are calibrated by bounded Levenberg–Marquardt on four SPX smiles (1–7 months), the four VIX futures and four VIX smiles. The modulation dynamics are fixed by the architecture. On 20 independent batches of 10 000 paths:

- SPX smiles: RMSE 0.63, 0.49, 0.51 and 1.12 vol points;
- VIX futures: within 0.18 points;
- VIX smiles near the money ($|k| \leq 0.3$): RMSE 3.4, 1.5, 3.0 and 3.4 vol points at 1, 2, 4 and 5 months;
- VIX smiles over all strikes: RMSE 7.3, 10.1, 4.7 and 4.1 vol points;
- VIX slopes: 1.02, 0.67, 0.41, 0.33 (market 1.21, 0.86, 0.54, 0.45).

On a single run of 40 000 independent paths, $Q_{\text{val}} = 4.56$, against $Q_{\text{cal}} = 3.04$.

The VIX smiles are fitted near the money at every expiry. Their right wings at 2–5 months are too flat: the modulation dynamics, selected on time series, give too little vol-of-vol convexity for option prices.

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1 Market data and targets

Two files of the same date (reference 19 February 2026) give European option quotes as Black-76 implied volatilities (bid and ask), with spot, forward and year fraction to expiry (ACT/365):

- **SPX**: spot 6 877.65; 11 expiries from 1 day to 5.8 years; implied volatilities on the quoted forward.
- **VIX**: spot 19.34; 4 expiries with futures 20.10, 20.73, 21.19, 21.72; implied volatilities on the VIX future.

Out-of-the-money quotes with both bid and ask are used (puts below the forward, calls above), subsampled evenly in log-moneyness $k = \ln(K/F)$:

Table 1: Calibration targets: expiry T (years), number of strikes, log-moneyness range. The quoted mid is the target.

market	T	strikes	k range
SPX	0.079	25	[-0.18, 0.07]
SPX	0.156	25	[-0.25, 0.09]
SPX	0.326	25	[-0.36, 0.13]
SPX	0.578	25	[-0.48, 0.17]
VIX	0.074	20	[-0.26, 0.91]
VIX	0.151	20	[-0.29, 0.98]
VIX	0.323	20	[-0.28, 0.95]
VIX	0.419	20	[-0.25, 0.93]

2 The model in continuous time

Time is in years. On $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{Q})$:

- $W^1, \dots, W^M, W^\perp$ are independent Brownian motions, and $W = (W^1, \dots, W^M)^\top$;
- L is an independent compound Poisson process with intensity Λ and $\mathrm{Exp}(a_J)$ jump sizes.

Layer 0′: vol-of-vol modulation.

$$d\Gamma_t = -\lambda_\Gamma \Gamma_t \, dt + dL_t, \qquad \Gamma_0 > 0, \qquad \bar{\Gamma} = \frac{\Lambda}{a_J \lambda_\Gamma}. \tag{2.1}$$

Layer 1: modulated rough bank. With $\lambda_i = \lambda_{lo}(\lambda_{hi}/\lambda_{lo})^{(i-1)/(M-1)}$,

$$dr_t^i = -\lambda_i r_t^i\,dt + \sqrt{2\lambda_i\Gamma_t}\,dW_t^i, \qquad \zeta_t = \frac{q^\top r_t}{\sqrt{\gamma_0}}, \tag{2.2}$$

$$q_i = \sqrt{b_i}, \qquad b = \arg\min_{b\geq 0} \int_{\tau_{\min}}^{\tau_{\max}} \Big(\frac{2\sum_i b_i(1-e^{-\lambda_i\tau})}{\tau^{2H}} - 1\Big)^2 \frac{d\tau}{\tau}, \qquad \gamma_0 = \textstyle\sum_i b_i.$$

Layer 0: return noise and leverage.

$$dW_t^X = \varrho\, u^\top dW_t + \sqrt{1-\varrho^2}\,dW_t^\perp, \qquad u = \frac{m}{\|m\|}, \quad m_i = q_i\sqrt{2\lambda_i}\,\frac{1-e^{-\lambda_i\Theta}}{\lambda_i\Theta}.$$

Layer 2: endogenous feedback bank. For $j = 1, \dots, J$,

$$ds_t^j = -\kappa_j s_t^j\,dt + \sqrt{2\kappa_j}\,\frac{\sigma_t}{\tilde\sigma(t)}\,dW_t^X, \qquad P_t = w^\top s_t, \qquad \kappa_j = \kappa_{hi}\Big(\frac{\kappa_{lo}}{\kappa_{hi}}\Big)^{\frac{j-1}{J-1}}, \quad \|w\| = 1.$$

Readout, volatility and price.

$$\sigma_t = \tilde\sigma(t)\,e^{\psi(\zeta_t)}\,F(P_t), \qquad dX_t = -\tfrac12\sigma_t^2\,dt + \sigma_t\,dW_t^X, \qquad S_T = F_T\,e^{X_T}, \tag{2.3}$$

with F_T the quoted forward, and

$$\tilde\sigma(t) = a\,e^{-bt} + c\,(1-e^{-bt}). \tag{2.4}$$

Definition 2.1 (Readouts)**.** 1. ψ is continuous and piecewise linear on the $N_c{+}1 = 6$ equispaced knots $-3, -1.8, -0.6, 0.6, 1.8, 3$. It has slope $-c_m \leq 0$ on the m-th piece, is normalised by $\psi(0) = 0$, and is constant beyond ± 3.

2. F is continuous and piecewise linear on the knots $-8, -5, -3, -1.5, 0, 5$ ($N_d = 4$ loss-side pieces, $N_e = 1$ gain-side piece). It satisfies $F(0) = 1$, $F(p_j) = F(p_{j+1}) + d_{j+1}$ on the loss side ($d \geq 0$) and $F(5) = e^{-e_1}$ on the gain side ($e_1 \geq 0$), and is constant beyond the outer knots.

Both readouts are non-increasing and bounded. The numbers of pieces and the knots are those of the grid-search optimum.

Today's state. Given (ζ_0, Γ_0, P_0) and $g \sim \mathcal{N}(0, I_M)$,

$$r_0 = \sqrt{\Gamma_0}\,g - q\,\frac{q^\top\sqrt{\Gamma_0}g - \zeta_0\sqrt{\gamma_0}}{\|q\|^2}, \qquad s_0 = P_0 w,$$

i.e. the stationary law at level Γ_0, conditioned on $q^\top r_0 = \zeta_0\sqrt{\gamma_0}$.

VIX. With $\tau = 30/365$,

$$\mathrm{VIX}_T = 100\sqrt{\frac{1}{\tau}\,\mathbb{E}^{\mathbb{Q}}\Big[\int_T^{T+\tau}\sigma_u^2\,du\,\Big|\,\mathcal{F}_T\Big]}, \tag{2.5}$$

with futures $\mathbb{E}^{\mathbb{Q}}[\mathrm{VIX}_T]$ and options $\mathbb{E}^{\mathbb{Q}}[(\mathrm{VIX}_T - K)^{\pm}]$, undiscounted as quoted.

Table 2: Fixed architecture: optimum of the grid-search protocol (rates in yr^{-1}).

M	18	J	2
H	0.0577	κ_{lo}, κ_{hi}	1.008, 80.22
λ_{lo}, λ_{hi}	0.001, 231.9	w	(0.995, -0.099)
Θ (days)	36.8	γ_0	0.679
λ_Γ, Λ	2.467, 4.628	a_J (mean jump $1/a_J$), $\bar\Gamma$	1.063 (0.941), 1.765
(N_c, N_d, N_e)	(5, 4, 1)		

Proposition 2.2 (Well-posedness and moments)**.** *Under Definition 2.1,*

$$\sigma_t \le \sigma_{\max} := \max_t \tilde{\sigma}(t)\, e^{\max \psi} \max F < \infty.$$

Consequently:

1. *the system* (2.1)–(2.3) *has a unique strong solution;*
2. *X, P and VIX_T have finite moments of all orders;*
3. *e^X is a true martingale, so $\mathbb{E}^{\mathbb{Q}}[S_T] = F_T$.*

Proof. Γ is explicit and positive. r solves a linear SDE driven by $\sqrt{\Gamma}$ and has finite moments, because Γ has exponential tails. Given (Γ, r), the pair (s, X) solves an SDE whose coefficients are Lipschitz in s and bounded. Bounded σ implies Novikov's condition, hence the martingale property, and bounds every moment of X, P and $\int \sigma^2$. □

3 Pricing

3.1 Simulation

The model is simulated with one step per calendar day ($\Delta t = 1/365$) up to the horizon 0.6:

- **Layer 1**: the exact bivariate law of $(\Delta W^i, \int e^{-\lambda_i(\cdot)} dW^i)$, scaled by $\sqrt{\Gamma}$ frozen over the step;
- Γ: exact decay $e^{-\lambda_\Gamma \Delta t}$ plus the pre-drawn jumps of Section 3.2;
- **Layer 2**: exponential integration;
- X: Euler step.

SPX payoffs use the weighted martingale correction $S_T = F_T e^{X_T} / \sum_i \omega_i e^{X_T^i}$, with ω the normalised importance weights. Implied volatilities are obtained by vectorised bisection on the Black-76 price. The architecture's rates are used in the same units as in the grid search (per year), and the time step is calendar time.

3.2 Importance sampling of the jumps

Proposition 3.1 (Likelihood ratio)**.** *Let the jumps of L on $[0, H]$ be drawn under a reference law with intensity Λ_{ref} and $\mathrm{Exp}(a_{\mathrm{ref}})$ sizes. For any functional f of the path on $[0, H]$,*

$$\mathbb{E}[f] = \mathbb{E}_{\mathrm{ref}}[f\, w], \qquad w = \Big(\frac{\Lambda}{\Lambda_{\mathrm{ref}}}\Big)^N e^{-(\Lambda - \Lambda_{\mathrm{ref}})H} \prod_{j=1}^{N} \frac{a_J}{a_{\mathrm{ref}}} e^{-(a_J - a_{\mathrm{ref}})z_j},$$

where N is the number of jumps and z_j their sizes.

Proof. w is the Radon–Nikodym derivative of the law of a marked Poisson process with intensity Λ and mark density $a_J e^{-a_J z}$ with respect to the reference law. The jump times have the same conditional (uniform) law under both, so they do not enter. □

The implementation uses $H = 0.6$ and:

- stratified counts, $N_i = F^{-1}_{\mathrm{Poi}(\Lambda_{\mathrm{ref}} H)}\big((\pi(i) + U_i)/n\big)$;
- uniform jump times;
- self-normalised weighted averages.

Note. Here $(\lambda_\Gamma, \Lambda, a_J) = (2.47, 4.63, 1.06)$ are fixed by the architecture, and the reference law is set equal to the model law: $\Lambda_{\mathrm{ref}} = \Lambda$, $a_{\mathrm{ref}} = a_J$. All weights then equal 1 and the effective sample size is the full number of paths, while the counts stay stratified. A reference law far from the model law makes the likelihood ratios very dispersed: with $\Lambda_{\mathrm{ref}} = 3$ and mean size 3 the effective sample size would be about 180 of 25 000 paths.

3.3 The VIX estimator

Lemma 3.2 (Conditional law of the rough factor)**.** *Let* $\Gamma_{T+s} \approx \Gamma_T e^{-\lambda_\Gamma s} + \bar{\Gamma}(1 - e^{-\lambda_\Gamma s})$*. Then, given* (r_T, Γ_T)*,* ζ_{T+h} *is Gaussian with*

$$m_h = \frac{1}{\sqrt{\gamma_0}} \sum_i q_i e^{-\lambda_i h} r_T^i, \qquad v_h = \frac{1}{\gamma_0} \sum_i q_i^2 \Big[\Gamma_T \, c_{1,i}(h) + \bar{\Gamma} \, c_{2,i}(h)\Big],$$

$$c_{1,i}(h) = \frac{2\lambda_i \big(e^{-\lambda_\Gamma h} - e^{-2\lambda_i h}\big)}{2\lambda_i - \lambda_\Gamma}, \qquad c_{2,i}(h) = 1 - e^{-2\lambda_i h} - c_{1,i}(h).$$

Proof. $r^i_{T+h} = e^{-\lambda_i h} r^i_T + \int_T^{T+h} \sqrt{2\lambda_i \Gamma_u}\, e^{-\lambda_i (T+h-u)} dW^i_u$. Given Γ, the integral is Gaussian with variance $\int_0^h 2\lambda_i \Gamma_{T+s} e^{-2\lambda_i (h-s)} ds$. Substituting the conditional mean of Γ and integrating gives $c_{1,i}$ and $c_{2,i}$. □

Proposition 3.3 (Closed form for the readout)**.** *If* $\psi = \alpha_j + \beta_j z$ *on* (a_j, b_j)*, with* $\beta = 0$ *on the outer pieces, and* $\zeta \sim \mathcal{N}(m, v)$*, then*

$$\mathbb{E}\big[e^{2\psi(\zeta)}\big] = \sum_j e^{2\alpha_j + 2\beta_j m + 2\beta_j^2 v} \Big[\Phi\Big(\frac{b_j - m - 2\beta_j v}{\sqrt{v}}\Big) - \Phi\Big(\frac{a_j - m - 2\beta_j v}{\sqrt{v}}\Big)\Big].$$

Proof. Complete the square: $e^{2\beta z}$ times the $\mathcal{N}(m, v)$ density equals $e^{2\beta m + 2\beta^2 v}$ times the $\mathcal{N}(m + 2\beta v, v)$ density. □

Definition 3.4 (Positive VIX estimator)**.** For each VIX expiry T and each path, the base is

$$b_T = \int_0^\tau \tilde{\sigma}(T+h)^2 \, \mathbb{E}\big[e^{2\psi(\zeta_{T+h})} \,\big|\, r_T, \Gamma_T\big] \, F\big(\bar{P}_{T+h}\big)^2 \, dh, \qquad \bar{P}_{T+h} = \sum_j w_j e^{-\kappa_j h} s_T^j,$$

computed with a 12-point midpoint rule. The realised variance of the same path is regressed, by weighted least squares, as a ratio to the base on $(1, P_T, P_T^2, \Gamma_T, P_T\Gamma_T)$, giving a multiplier m. Then

$$\mathrm{VIX}_T^2 = \frac{100^2}{\tau} \, b_T \cdot \mathrm{clip}(m, 0.1, 10).$$

4 Calibration

4.1 Parameters

The calibrated vector $\theta \in \mathbb{R}^{17}$ consists of:

- the readout slopes $c_{1..5}$ and the increments $d_{1..4}$, e_1;
- the term structure (a, b, c);
- ϱ;
- today's state (ζ_0, P_0, Γ_0).

The modulation dynamics $(\lambda_\Gamma, \Lambda, a_J)$ are *not* calibrated. In the current model they are hyperparameters of the rough bank, fixed with the rest of the architecture at the grid-search optimum. The bounds are given in Table 3.

4.2 Objective

The residual is a convex combination of blocks:

- four SPX smiles, total weight 0.45, tolerance max(half-spread, 0.003);
- the four VIX futures, weight 0.15, tolerance 0.10 points;
- four VIX smiles, total weight 0.40, tolerance max(half-spread, 0.015).

For block b with n_b terms,

$$r_{b,i} = \sqrt{w_b}\,\frac{\sqrt{\omega_{b,i}}\left(\mathrm{iv}^{\mathrm{mod}}_{b,i} - \mathrm{iv}^{\mathrm{mkt}}_{b,i}\right)}{s_{b,i}\sqrt{n_b}}, \qquad Q(\theta) = \|r(\theta)\|^2,$$

with $\omega \equiv 1$ for SPX and futures. For the VIX smiles the near-the-money weights are

$$\omega_i = \frac{e^{-k_i^2/(2h^2)}}{\frac{1}{n}\sum_j e^{-k_j^2/(2h^2)}}, \qquad h = 0.2.$$

A model price outside the no-arbitrage range contributes 5 tolerances.

4.3 Optimiser

Algorithm 1 Calibration

1: load targets (Table 1) and the architecture (Table 2)
2: $\theta_0 \leftarrow$ warm start (`results/warm_start.json`)
3: **Stage W (warm-up).** Bounded TRF (`x_scale="jac"`) on $n = 6\,000$ paths, at most 30 evaluations
4: **Stage C (main).** Bounded TRF from the end of stage W on $n = 25\,000$ paths, at most 12 evaluations
5: In both stages: common random numbers (Gaussian seed 2026, stratified reference jumps); forward-difference Jacobian (relative step 0.01) evaluated on 2 processes
6: **return** the best logged θ of stage C

Note. Stage W, on fewer paths, brings the model cheaply into the right region of the parameter space before the 25 000-path stage.

5 Results

5.1 Calibration

Stage W (6 000 paths) descends from $Q = 47.5$ to 4.9 in 385 evaluations. Stage C (25 000 paths) starts at $Q = 7.07$ and reaches $Q_{\mathrm{cal}} = 3.04$ in 236 evaluations, when its evaluation limit is reached, with the decrease slowing to below 1% per iteration.

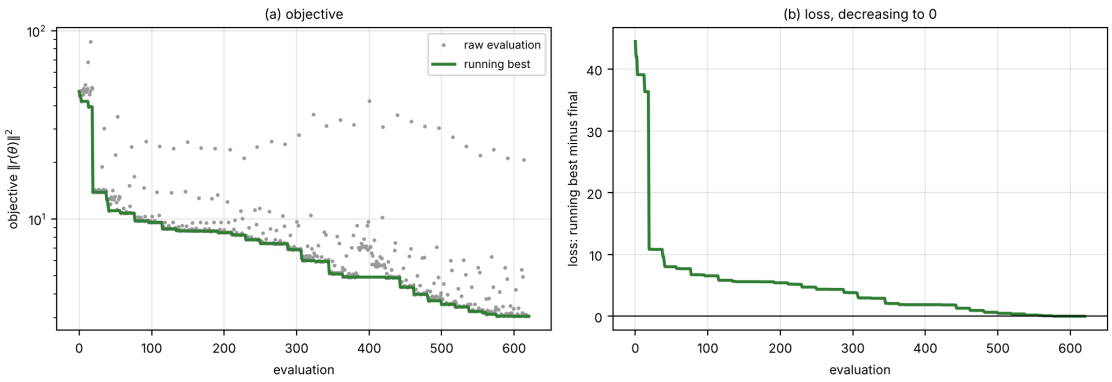


Figure 1: Calibration: objective per evaluation (stages W and C) and running best; the loss (running best minus final) decreases to 0.

5.2 Calibrated parameters

Table 3: Calibrated parameters and bounds. None ends on a bound.

parameter	role	bounds	value
c_1	slope of $-\psi$	[0.005, 3]	0.0076
c_2	slope of $-\psi$	[0.005, 3]	0.4701
c_3	slope of $-\psi$	[0.005, 3]	0.8317
c_4	slope of $-\psi$	[0.005, 3]	0.1547
c_5	slope of $-\psi$	[0.005, 3]	0.6080
d_1	increment of F (loss side)	[0, 6]	4.3497
d_2	increment of F (loss side)	[0, 6]	1.8100
d_3	increment of F (loss side)	[0, 6]	1.7007
d_4	increment of F (loss side)	[0, 6]	3.3470
e_1	log-decay of F (gain side)	[0, 3]	1.5620
a	level today	[0.02, 0.8]	0.3558
b (yr^{-1})	speed	[0.1, 60]	5.1086
c	long-run level	[0.02, 0.8]	0.1099
ϱ	leverage correlation	[0, 0.999]	0.9264
ζ_0	rough state today	$[-6, 6]$	5.3534
P_0	feedback state today	$[-8, 8]$	-0.6067
Γ_0	Γ today	[0.02, 8]	0.0993

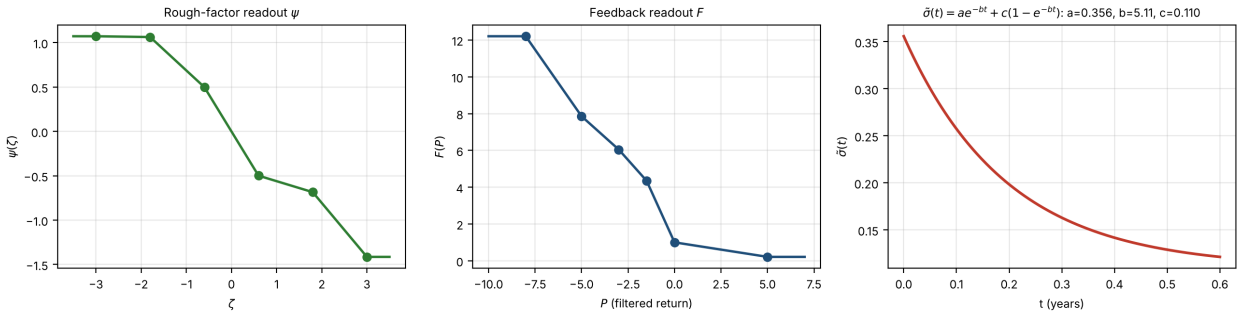


Figure 2: Calibrated readouts ψ and F, and $\tilde{\sigma}(t)$.

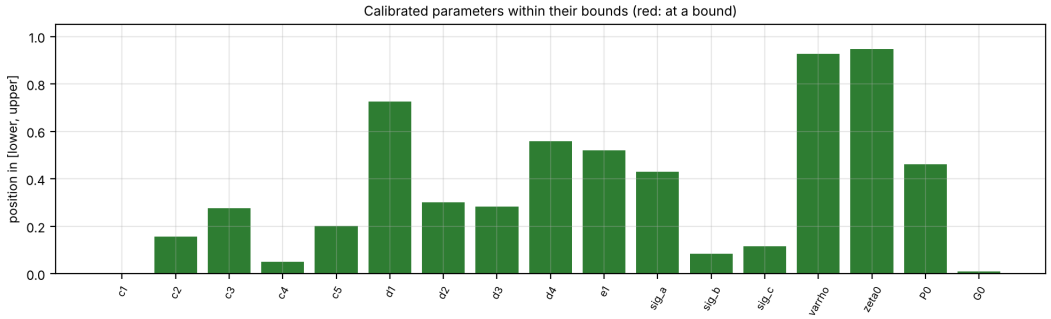


Figure 3: Calibrated parameters within their bounds.

- **Readout ψ.** It is flat for $\zeta \le -1.8$ (cap $e^{1.07} \approx 2.9$) and steep on $[-0.6, 0.6]$. It keeps falling for $\zeta > 0$, down to $e^{-1.41} \approx 0.24$.
- **Feedback F.** It is very strong on the loss side ($F(-1.5) = 4.3$, $F(-8) = 12.2$) and falls to 0.21 after large gains. Together with $\varrho = 0.93$, it carries most of the SPX skew.
- **Term structure.** $\tilde{\sigma}$ decreases from 0.356 to 0.110, with a half-life of 1.6 months.

- **Today's state.** $\zeta_0 = 5.35$ is high, and $\Gamma_0 = 0.10$ is far below the stationary mean $\bar{\Gamma} = 1.77$. Today's vol-of-vol is low and rises in expectation over the VIX horizons, which is what fits the short VIX smile.

5.3 Validation on independent paths

All results below use random numbers independent of the calibration. The main validation re-prices the model on 20 independent batches of 10 000 paths (seeds 1000–1019).

Table 4: VIX smiles (mean of 20 batches): RMSE over $|k| \leq 0.3$ and over all strikes (vol points), linear slope in k (model and market), maximal 95% CI half-width, and fraction of strikes where the market mid lies in the CI.

T	RMSE $\lvert k\rvert \leq 0.3$	RMSE all	slope model	slope market	CI	mid in CI
0.074	3.4	7.3	1.02	1.21	±4.7	40%
0.151	1.5	10.1	0.67	0.86	±2.8	50%
0.323	3.0	4.7	0.41	0.54	±2.0	20%
0.419	3.4	4.1	0.33	0.45	±2.1	15%

Table 5: SPX smiles (mean of 20 batches): RMSE and maximal 95% CI half-width (vol points). Right: VIX futures (points).

T	RMSE	CI
0.079	0.63	±0.34
0.156	0.49	±0.30
0.326	0.51	±0.34
0.578	1.12	±0.30

T	market	model	95% CI
0.074	20.10	19.92	[19.80, 20.04]
0.151	20.73	20.66	[20.51, 20.82]
0.323	21.19	21.32	[21.20, 21.44]
0.419	21.72	21.82	[21.66, 21.98]

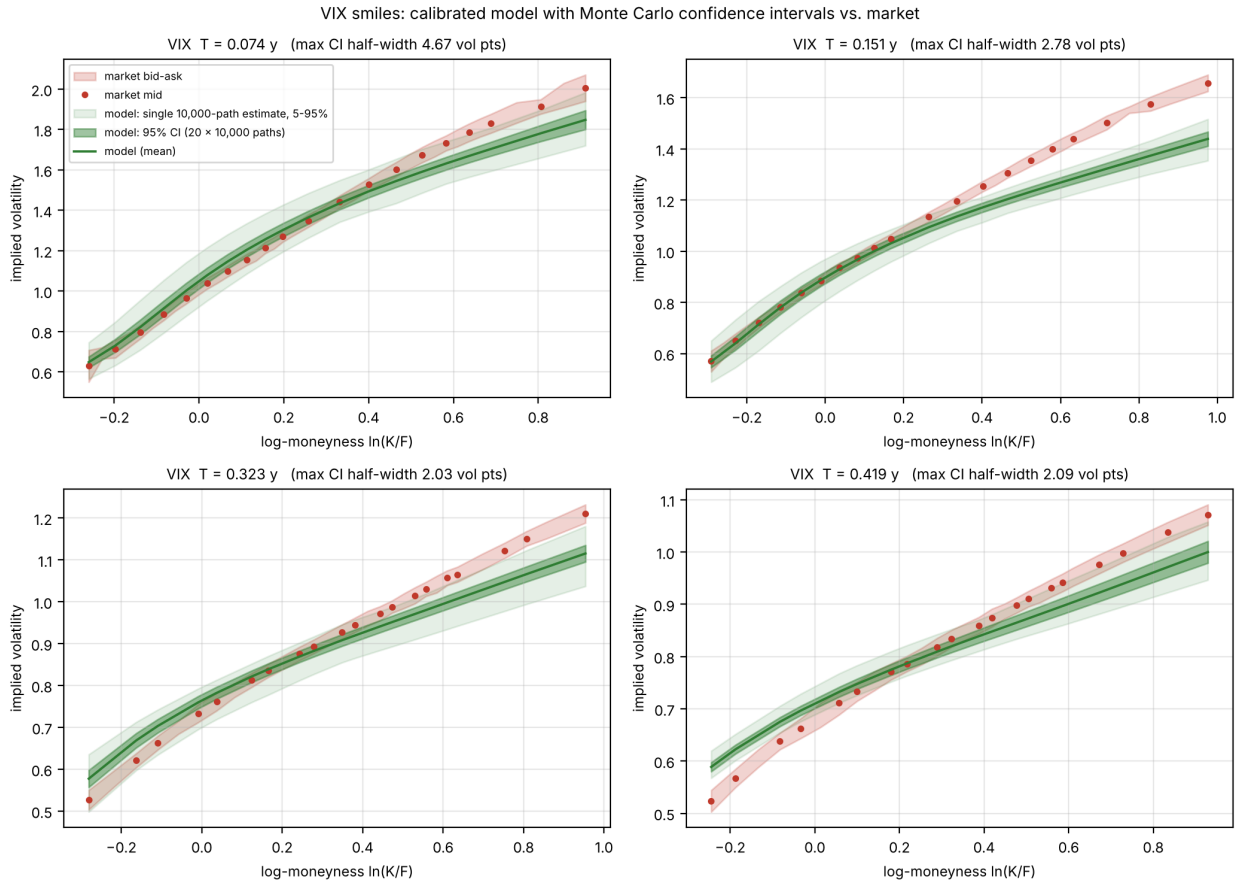


Figure 4: VIX smiles: model mean of 20 independent batches, 95% confidence interval, range of a single batch, and market bid–ask and mid.

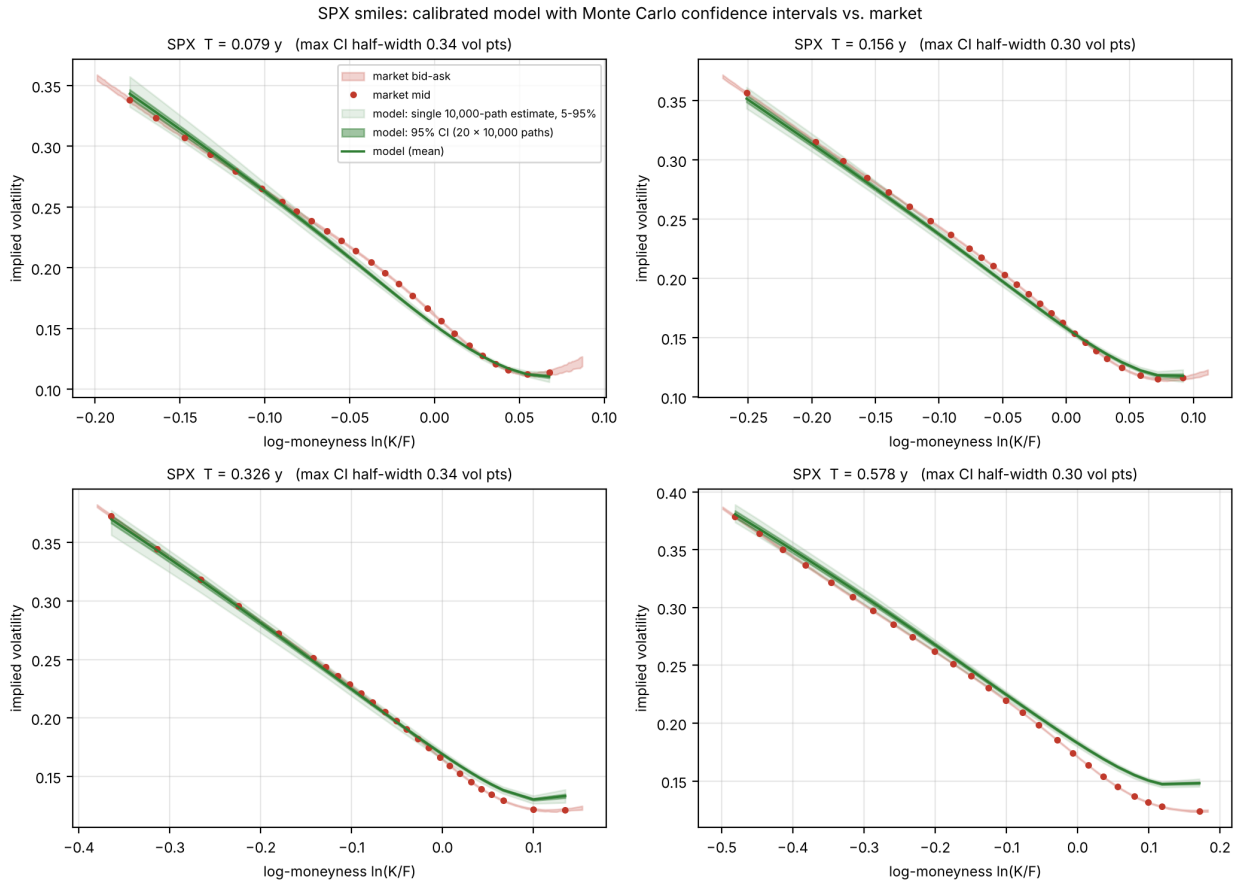


Figure 5: SPX smiles with Monte Carlo confidence intervals.

Single large run. On one run of 40 000 independent paths (seed 99), $Q_{\text{val}} = 4.56$, and the SPX RMSE is 0.65 / 0.31 / 0.54 / 1.07 vol points.

Table 6: VIX smile RMSE (vol points), single run of 40 000 independent paths.

T	$\lvert k\rvert \le 0.3$	all strikes
0.074	7.4	6.7
0.151	1.1	8.2
0.323	2.7	3.1
0.419	2.1	4.6

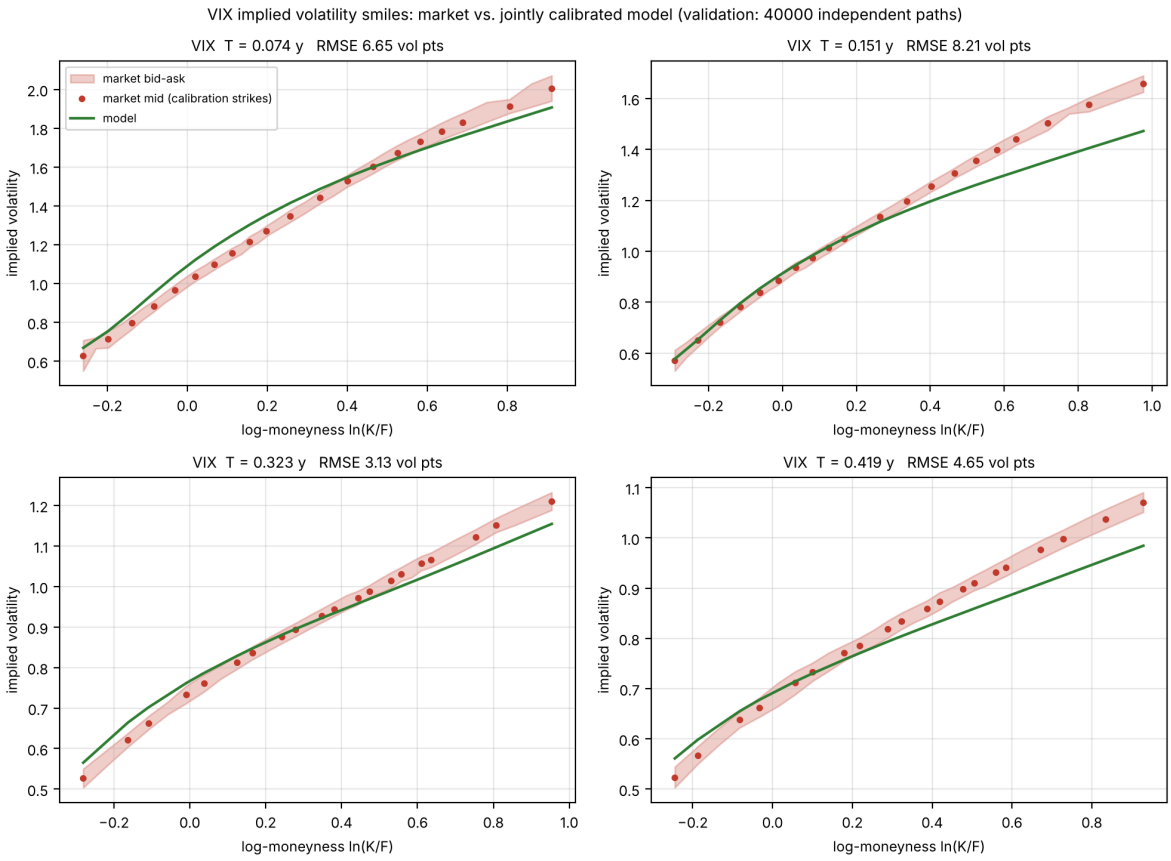


Figure 6: VIX smiles, single run of 40 000 independent paths.

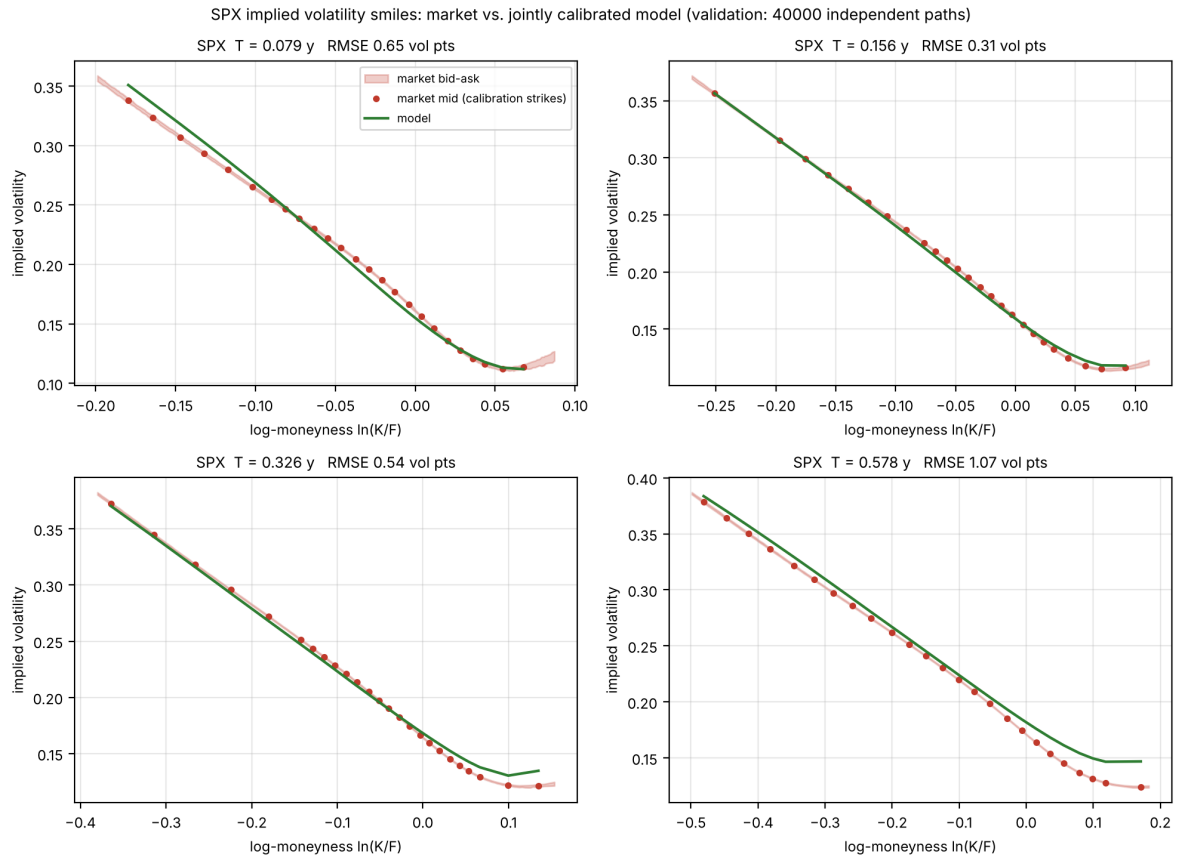


Figure 7: SPX smiles, single run of 40 000 independent paths.

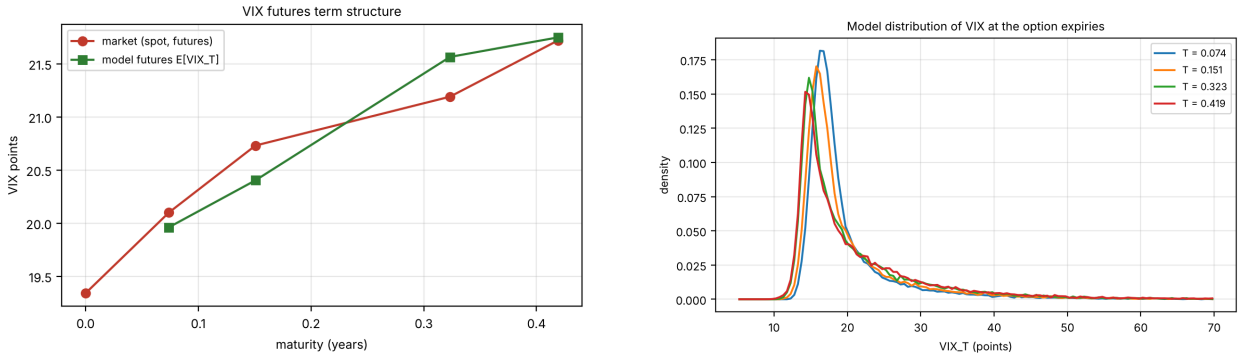


Figure 8: VIX futures term structure (left) and model densities of VIX_T (right).

5.4 Summary

- **VIX near the money.** RMSE 1.5–3.4 vol points at all four expiries.
- **VIX wings.** The model's smiles are too flat at 2–5 months: slopes 0.67/0.41/0.33 against market 0.86/0.54/0.45. The right wing is 10–20 vol points too low at 2 months, and the left wing slightly too high at 4–5 months.
- **SPX.** RMSE 0.49–1.12 vol points; the 7-month smile is the weakest.
- **VIX futures.** Within 0.18 points of the market.

6 Limitations

1. **Modulation dynamics fixed by the time series.** $(\lambda_\Gamma, \Lambda, a_J)$ were selected on daily stylised facts under ¶. The VIX right wing asks for rarer and larger jumps of Γ under $\mathbb{Q}$. Recalibrating these three parameters under $\mathbb{Q}$, i.e. allowing a risk premium on the vol-of-vol jumps, is the natural next step.

2. **Strong leverage and feedback.** $\varrho = 0.93$ and $F(-8) = 12.2$ are large. Under ¶ (joint calibration of the time series) the same model had $\varrho \leq 0.7$ and $F(-8) \leq 2.2$. The two measures ask for different readouts.

3. **Today's state** $\zeta_0 = 5.35$ is far in the tail. The initial state, three scalars, carries the short end.

4. **VIX estimator.** It is a regression approximation; a nested Monte Carlo check is still to be done.

5. **Calibration.** It was stopped at the evaluation limit, with the decrease below 1% per iteration. Its 25 000 paths with common random numbers leave a gap between $Q_{\text{cal}} = 3.04$ and $Q_{\text{val}} = 4.56$.

A Code reference

All modules carry a banner docstring. Run `run_mod.sh` from inside `code/`. It sources `env.sh` (Table 7), calibrates from `results/warm_start.json`, validates and exports the JSON results. `DATA_DIR` must contain `SPX_data_19_02_2026.csv` and `VIX_data_19_02_2026.csv`.

Table 7: Environment of the model (`env.sh`).

variable	value	meaning
`MODUL`, `MODUL_FIXED`	1, 1	modulated rough bank; $(\lambda_\Gamma, \Lambda, a_J)$ read from `architecture.json`, only Γ_0 calibrated
`FLEX_KNOTS`	$-3, -1.8, \ldots, 3$	knots of ψ ($N_c = 5$)
`FLEXF_KNOTS`	$-8, -5, -3, -1.5, 0, 5$	knots of F ($N_d = 4$, $N_e = 1$)
`JUMP_IS`, `LAM_REF`, `A_REF`	1, 4.628, 1.063	stratified jumps, reference law equal to the model law (weights 1)
`VIX_RATIO`	1	positive VIX estimator
`PSI_BOUNDED`	1	ψ constant beyond its knots
`SIGMA_FORM`	`param`	$\tilde{\sigma}(t) = ae^{-bt} + c(1 - e^{-bt})$
`VIX_K_BW`, `ZETA_MAX`	0.2, 6	near-the-money bandwidth; bound on $\lvert\zeta_0\rvert$

file	content
`cs_core.py`	market selection, architecture (`load_architecture`), Black-76 and implied volatility, near-the-money weights, residual
`cs_flex.py`	ψ: knot values from the slopes (`psi_vals`, $\psi(0) = 0$ by interpolation when 0 is not a knot), `psi_eval`, `e2psi_gauss` (Proposition 3.3)
`cs_flexd.py`	the model: F (`F_vals`, `F_eval`), $\tilde{\sigma}(t)$, parameter access (`unpack`, `modul_of`, with the fixed modulation dynamics), stratified jumps (`is_jumps`), simulation, positive VIX estimator (`vix_from_sim`), SPX and VIX smiles (`model_smiles`), warm start on new knots (`from_flexF`)
`cs_calibrate_mod.py`	Algorithm 1: stages W and C, forward-difference Jacobian on `NWORK` processes, trace and result files
`cs_evaluate.py`	single 40 000-path validation (seed 99), figures 1–7
`cs_smile_ci.py`	20 independent batches of 10 000 paths, figures 9–10
`cs_export_final.py`	`summary.json`, `trace.json`, `market_selection.json`
`architecture.json`	the grid-search optimum
`env.sh`, `run_mod.sh`	environment and full pipeline (about 40 minutes on 2 cores)

Note. The modules contain switches from earlier model variants (`BETA`, `FLOOR`, `SIGMA_FORM=bucket`). They are off here and have no effect unless set.

B Results files

file	content
`calibration.json`	bounds, calibrated θ, knot values of ψ and F, Q, block contributions, settings, architecture
`trace.json`	every calibration evaluation (stages W and C): θ, Q, block contributions, model VIX futures
`evaluation.json`	single 40 000-path validation: per slice strikes, market and model implied volatilities, errors
`smile_ci.json`	20-batch validation: per strike model mean, 95% CI, single-batch range; VIX futures with CI
`summary.json`	parameters with bounds, calibration settings, all validation metrics
`market_selection.json`, `warm_start.json`	the quotes used as targets; the starting point